NATIONAL CENTRE OF ARTIFICIAL INTELLIGENCE

RAI-853: Advanced Robotics with ROS

Lec 02a: The World of Robotics

Instructor:

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Outline

- Introduction to Robotics
- Applications of Robotics

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- **Introduction to Robotics**
- **Applications of Robotics**

Why Do We Need Robots?

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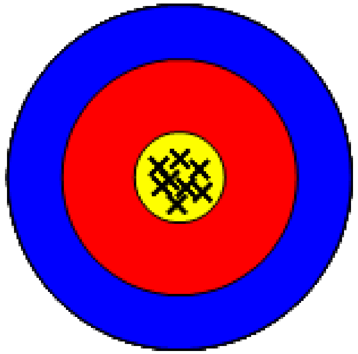
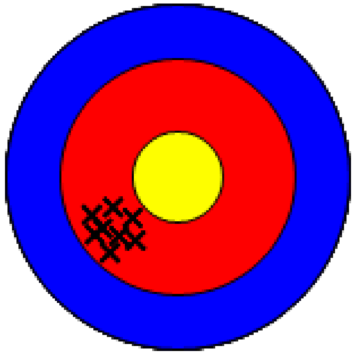
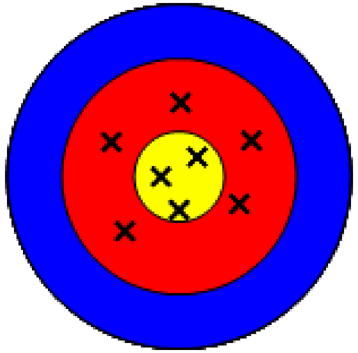
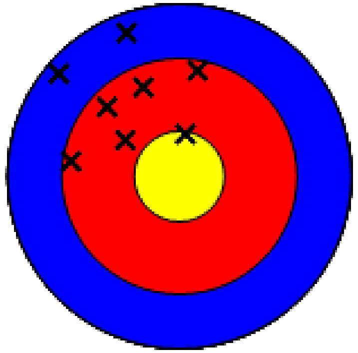
- Robots are used where the work is too labor intensive, boring or dangerous for a human being.

- The three D's of Robotics: Robots perform tasks that are considered **dull, dirty and dangerous** for human beings.

- Robots have a better accuracy, precision and work efficiency than human workers.

- Robots are Coooool ! ☺

Accuracy vs Precision

	Accurate	Inaccurate (systematic error)
Precise		
Imprecise (reproducibility error)		

What is a Robot?

What is a Robot?

- There is no standard definition of a Robot.

- Joseph Engelberger, who is considered as the Father of Robotics, famously said:

  "I can't define a robot, but I know one when I see one."

- As the field of Robotics keeps progressing, new definitions keep coming up.

What is a Robot?

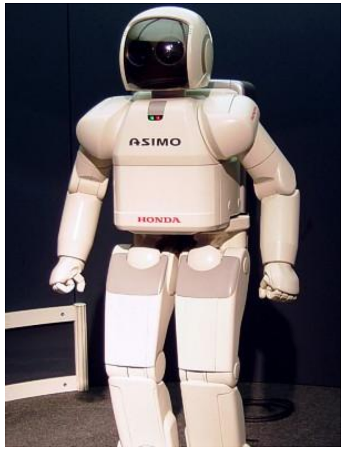
- Here is a definition that I like:

"Any automatically operated machine that replaces human effort, though it may not resemble human beings in appearance or perform functions in a humanlike manner."

(Encyclopedia Brittanica)

How do we differentiate between Robots and other machines?

Robots vs Machines

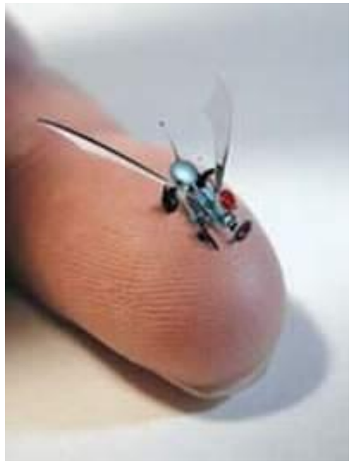


Characteristics of a Robot

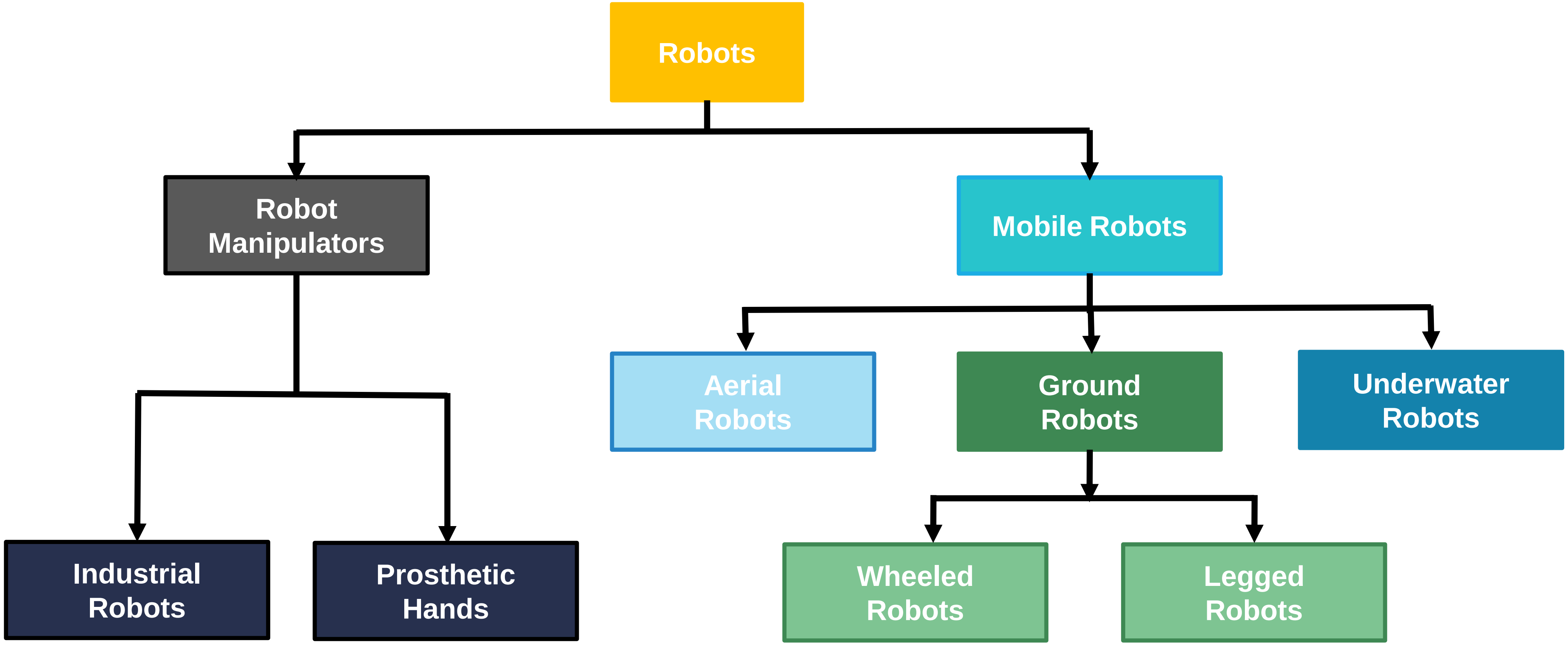
- Electromechanical System
- Reprogrammable
- Multifunctional
- Capable of Learning
- Capable of Sensing its Environment
- Capable of making decisions

Types of Robots

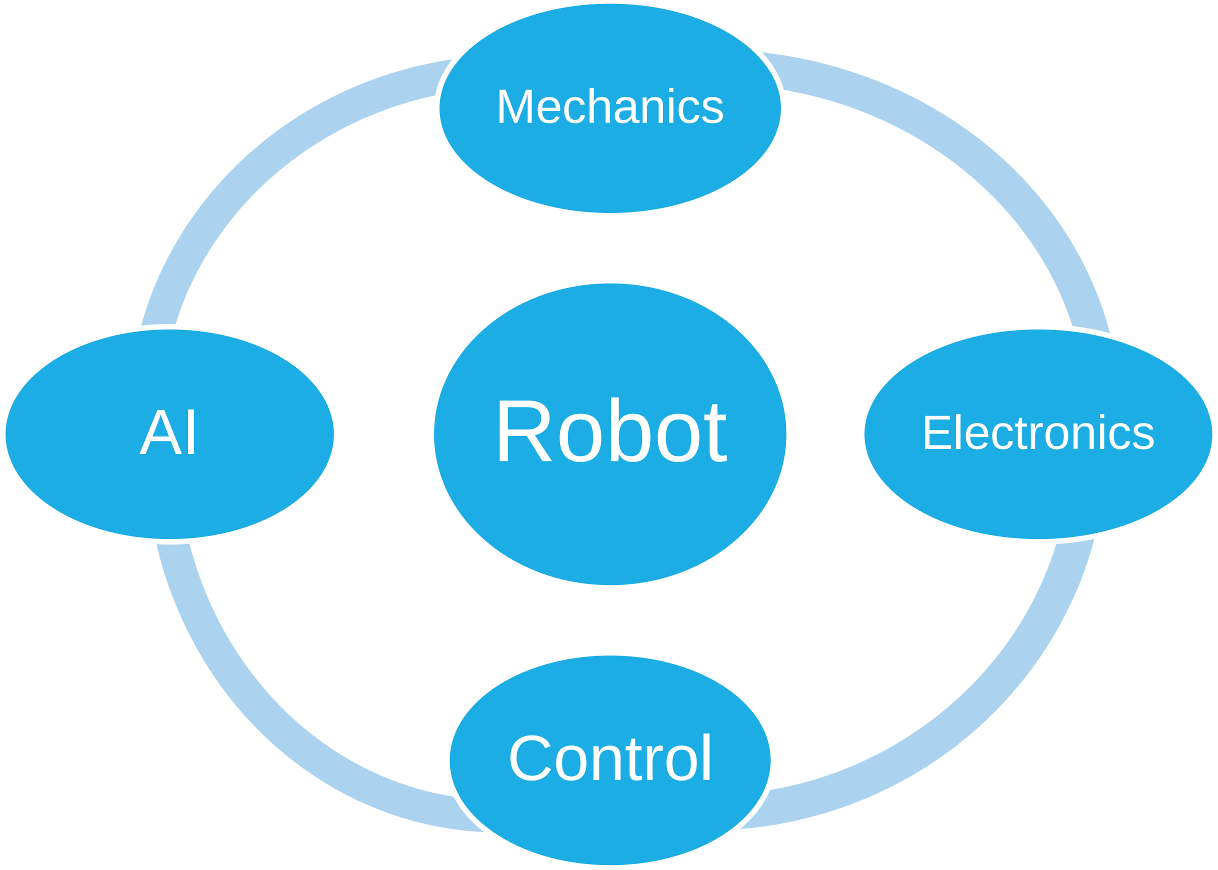
- Manipulators
- Wheeled Robots
- Humanoid Robots
- Multi Legged Robots
- Aerial Robots
- Micro Robots
- Underwater Robots



Types of Robots



Robotics is Multi Disciplinary



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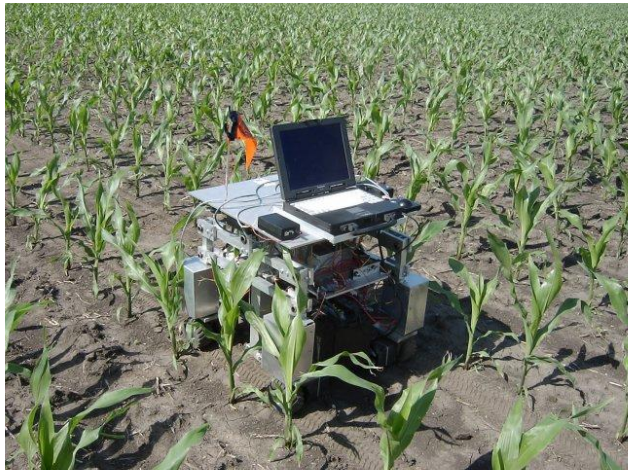
Applications of Robotics

Industrial Robots

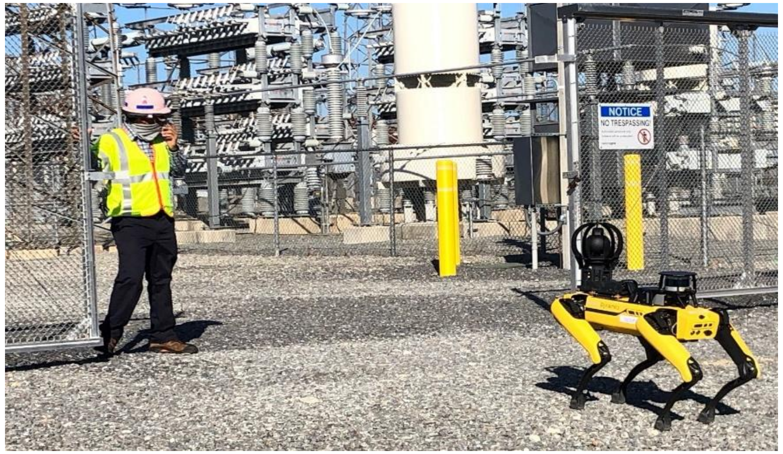
IRB 1200 - by ABB Robotics

Field Robots









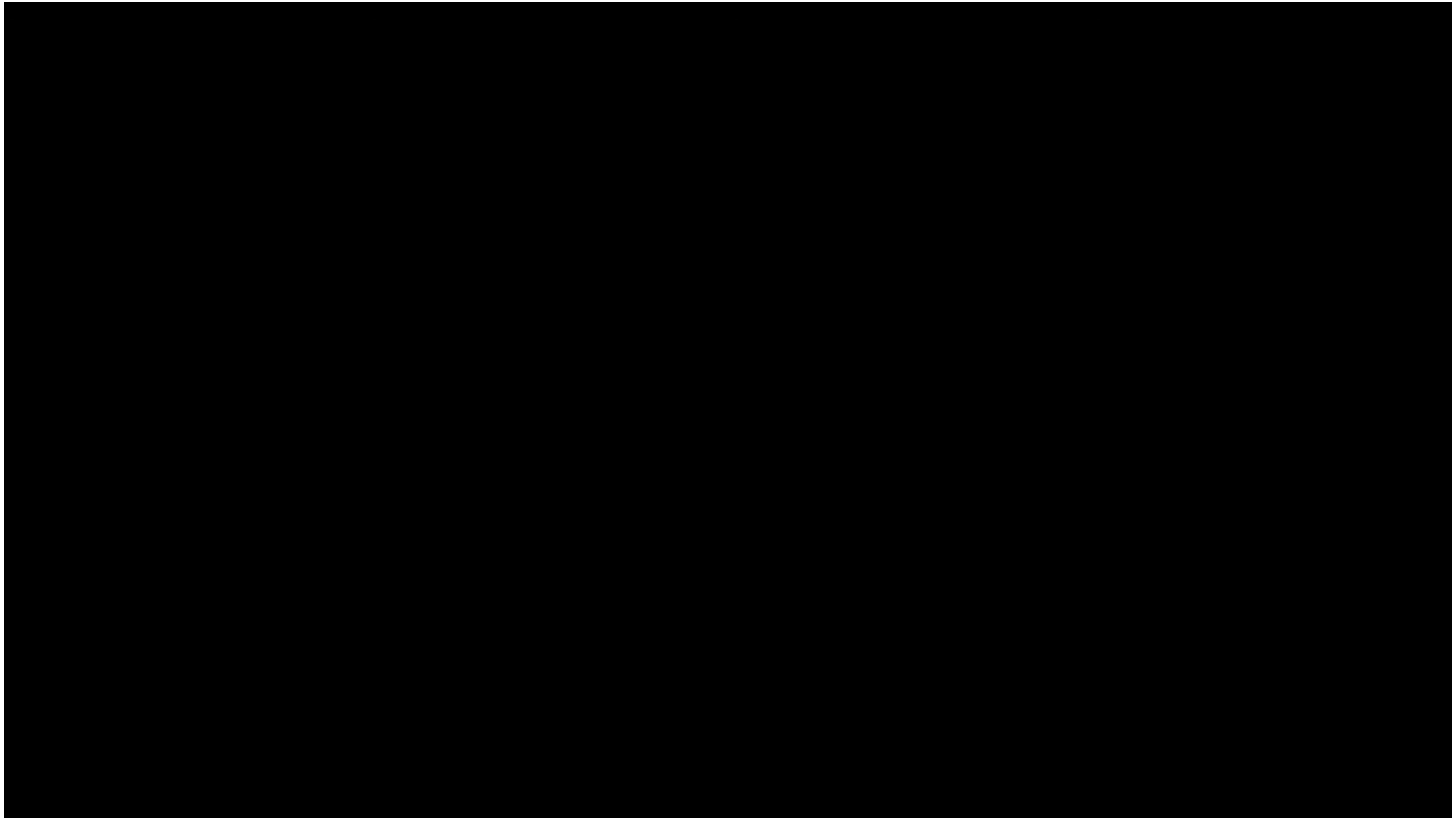
Autonomous Weeder by Carbon Robotics

Perseverance - Mars Rover

Applications of Robotics

Rehabilitation Robots

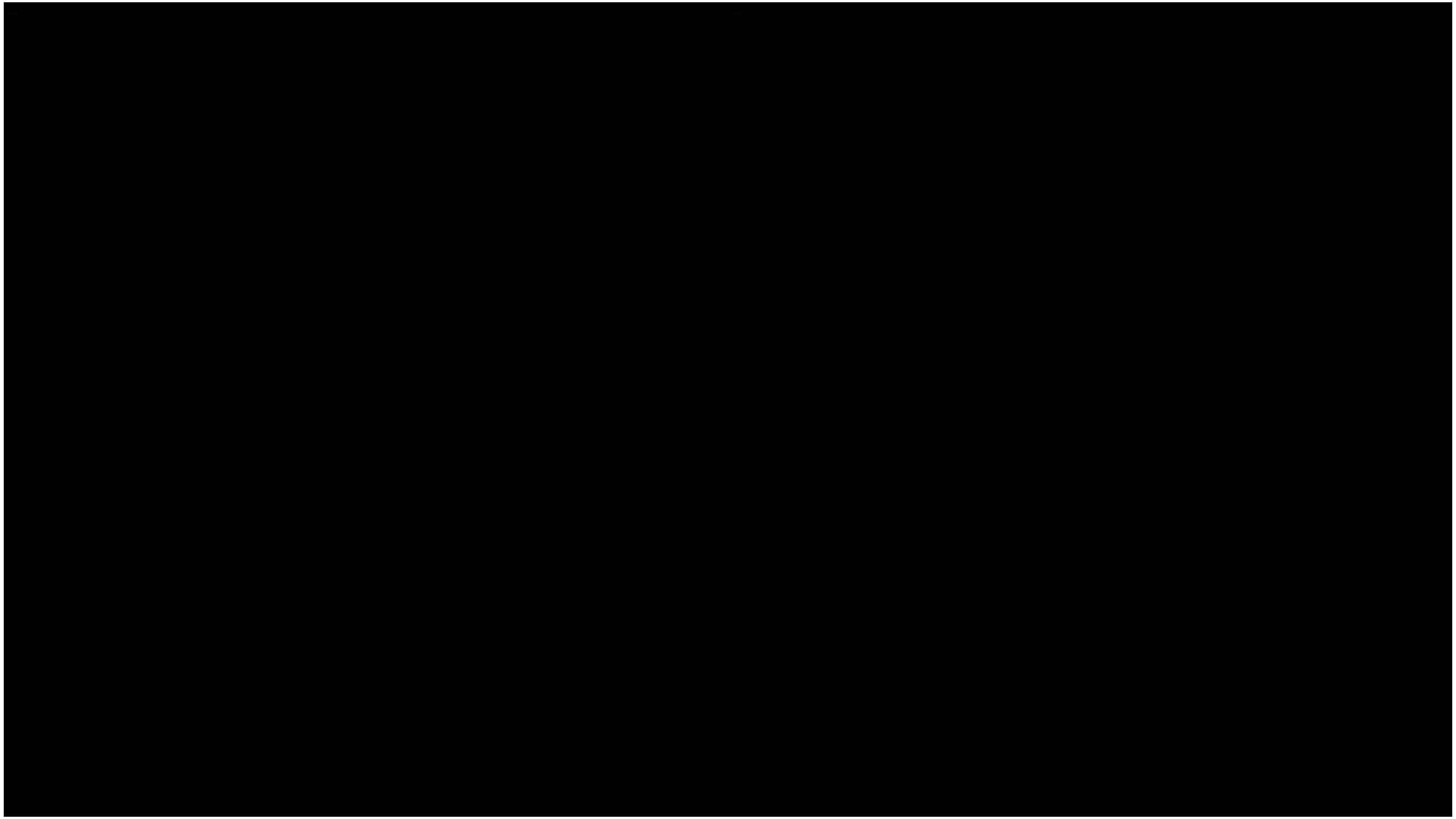
Exoskeleton – by Ekso Bionics



Applications of Robotics

Medical Robots

Da Vinci Surgical Robot



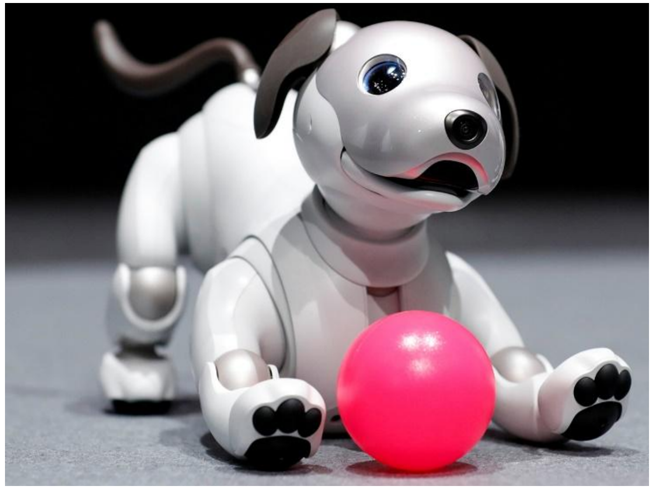
Domestic Robots

Kitchen Robot – by Moley Robotics

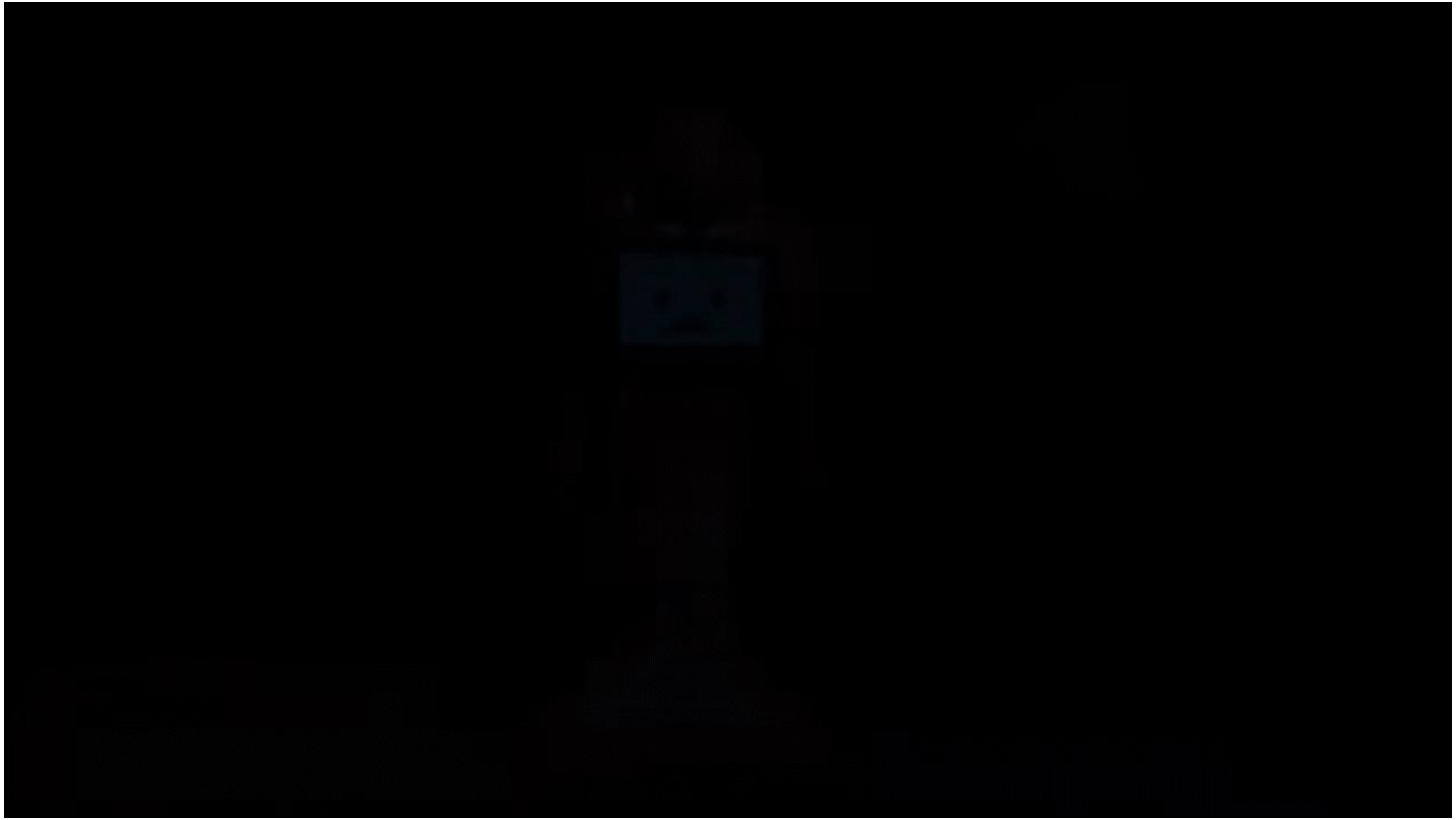
Digit – by Agility Robotics

Applications of Robotics

Entertainment Robots

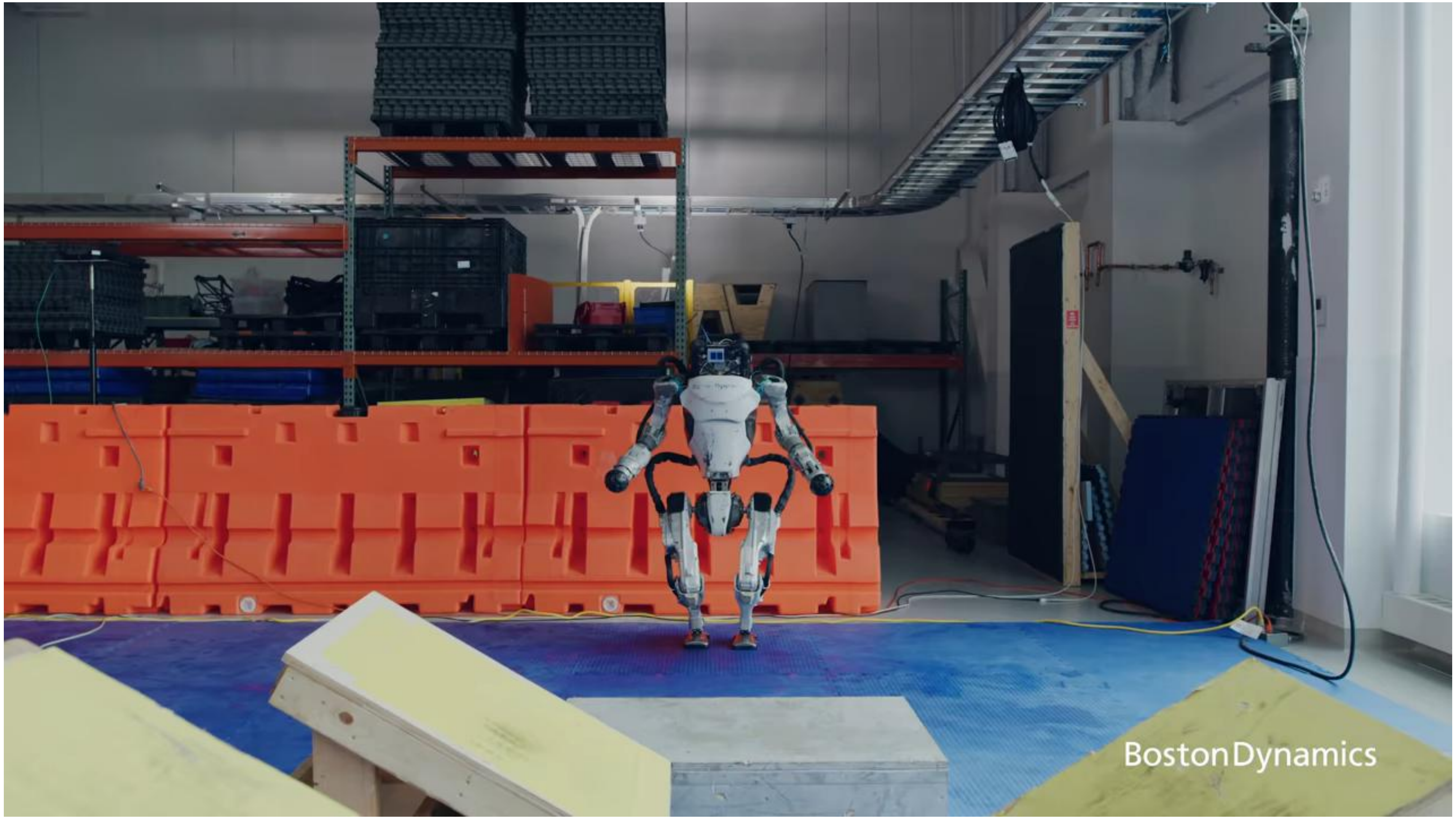


Pepper – by Softbank Robotics



Military Robots

Atlas – By Boston Dynamics



Challenges

- **Perception:**

  Extracting the “right” information from data.

- **Actuation:**

  Especially in Gripping and Walking.

- **Facing an unforeseen event:**

  Machine Learning only improves with more data.

- **Materials and Manufacturing:**

  Limiting factor in achieving better performance.

Dreams

- **Robotic Car:** Tesla, Google Car, DARPA Grand Challenge
- **Robot Maid:** Roomba, Scooba
- **Robot Surgeon:** da Vinci, ZEUS
- **Human Level Intelligence and Movement:** Robocup, DARPA Robotics Challenge
- **Robot Army:** It’s coming

Thank you for your attention.

Any Questions?